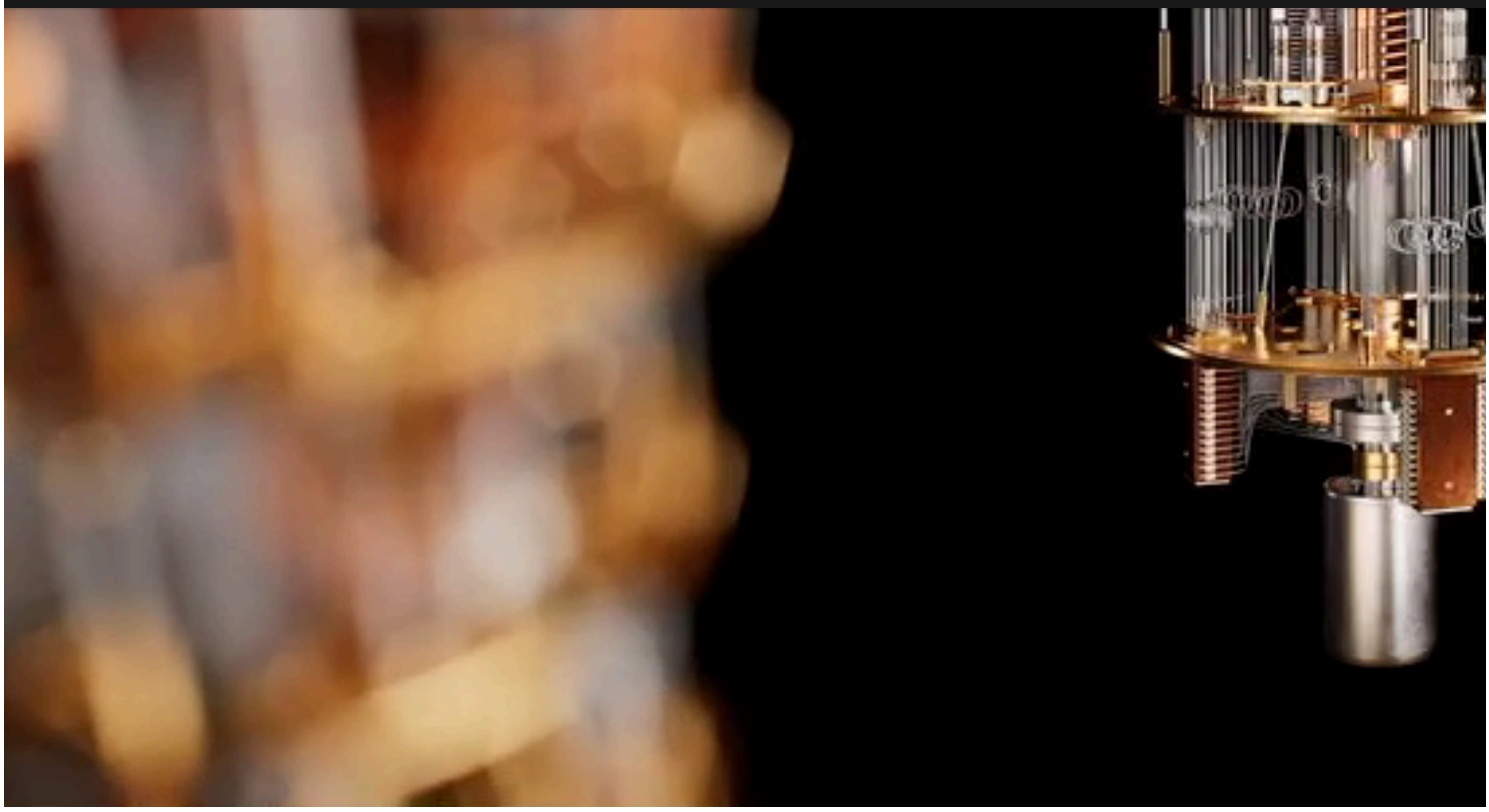
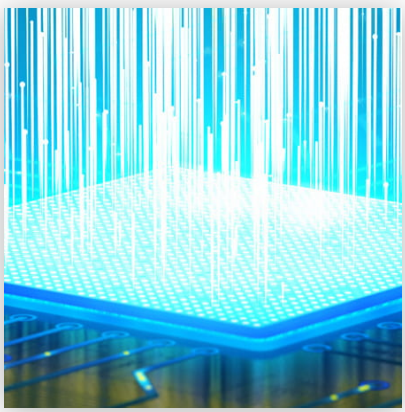
LPS | LABORATORY FOR PHYSICAL SCIENCES



SOLID STATE & QUANTUM PHYSICS

The Solid State and Quantum Physics (SSQP) research program conducts experimental and theoretical research focused on fundamental research and enabling technologies of quantum devices, cryogenic measurements, materials science, and associated information theory. Our research groups are active in the broad research community and collaborate with academia, industry, and national laboratories.

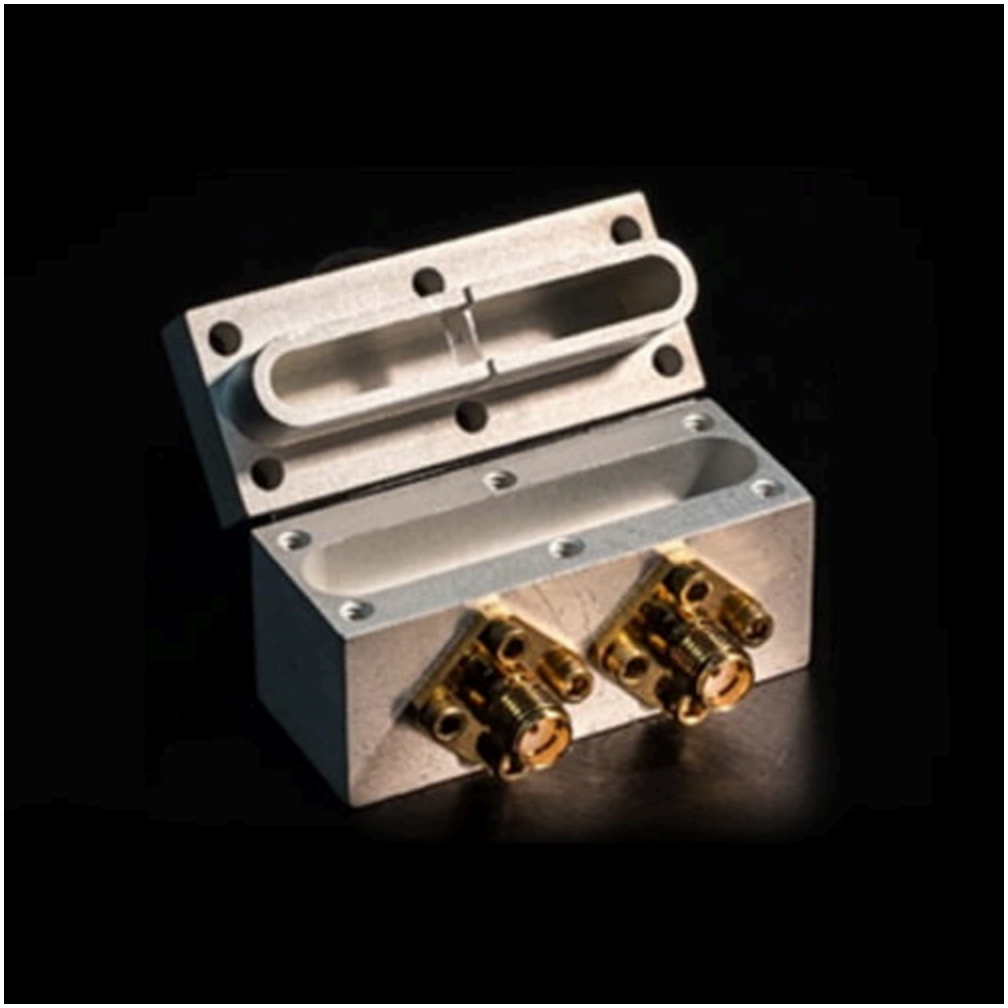


Since 1995, LPS has funded the quantum computing research community worldwide. LPS, in partnership with the Army Research Office, funds a broad array of quantum computing and related research, including these programs:

- HiPS (High Performance Superconducting Qubit Systems)
- QiS (Qubits in Silicon Program)
- SHiFT (Stable High Fidelity Trapped Ion Systems)
- QCISS (Quantum Characterization of Intermediate-Scale Systems)
- NEQST (New & Emerging Qubit Science & Technology)
- CQTS (Cross-Quantum Technology Systems)



The LPS Qubit Collaboratory (LQC) is a national Quantum Information Science Research Center hosted at the Laboratory for Physical Sciences at the University of Maryland, College Park. The LQC pursues disruptive qubit research, innovative workforce development programs, and deep, collaborative partnerships to tackle some of the hardest open problems in quantum information science and technology. **qubitcollaboratory.org**



For over 20-years LPS researchers have addressed fundamental challenges in quantum computing and solid-state physics. Our technical expertise and laboratory investments focus on the following research:

• Applied materials physics including epitaxial materials, atomistic fabrication, and defects in superconducting devices.
• Experimental quantum device fabrication and measurement using superconductor and semiconductor qubit devices.
• Quantum Information and device theory to propose new quantum devices and algorithms for quantum information science.

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